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Visual Grounding via Accumulated Attention

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Visual Grounding

Visual Grounding (VG) aims to locate the most relevant object or region in an image, from a given input query.

Query from GuessWhat?!: is it a person? Yes is the person a man? No is it the left woman? No does she wear glasses? Yes	**Query from ReferCOCO:** woman in black
	Query from ReferCOCO+: woman wearing glasses with smile
	Query from ReferCOCOg: a woman in black playing a game with her friends

Visual Grounding

The Visual Grounding problem can be decomposed into three sub-problems:

1. identify the main focus in the query
2. understand the concepts in the image
3. locate the most relevant object

Problems Transformation

We transform the VG problem into three attention problems:

1. which words to focus on in a query
2. where to look in an image
3. which object to locate

Proposed Architecture

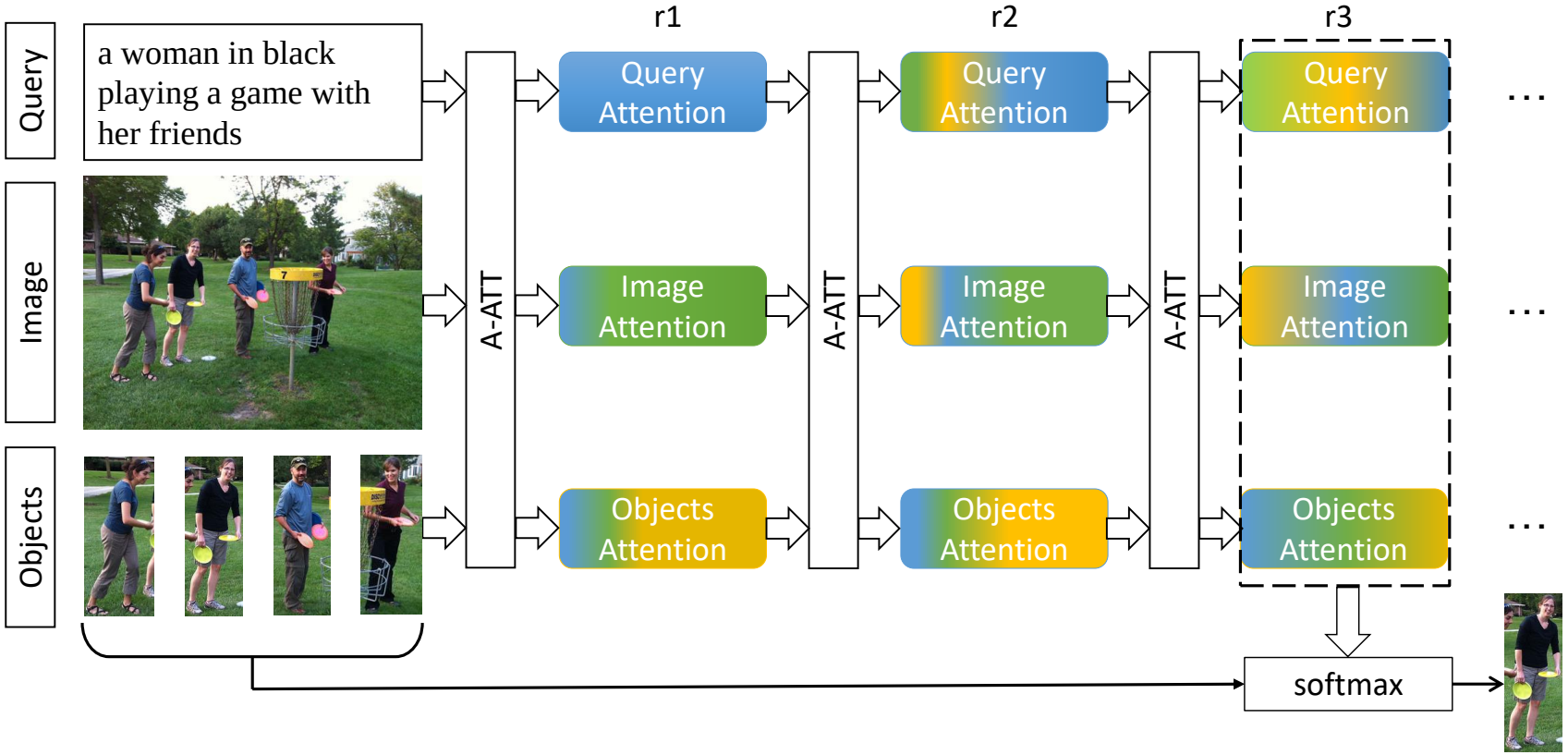


Figure: The proposed architecture.

Which words to focus on in a query

- given a dialogue consisting of a sequence of question-answer (QA) pairs $Q = \{\{q_1, a_1\}, \{q_2, a_2\}, ..., \{q_T, a_T\}\}$.
- use a hierarchical architecture to encode Q as $\boldsymbol{S} = \{\boldsymbol{s}_1, \boldsymbol{s}_2, ..., \boldsymbol{s}_T\}$
- compute the attention on the i-th QA pairs

$$\alpha_i^q = \text{softmax}(\boldsymbol{w}_q^\top \boldsymbol{H}_i^q), \quad i = 1, ..., T. \tag{1}$$

- $\boldsymbol{H}_i^q$ is a joint feature generated by the proposed accumulate operation

Where to look in an image

- employ the VGG-16 model to extract feature maps $\boldsymbol{V} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_L\}$ from the last convolutional layer.
- compute the attention on the i-th region.

$$\alpha_i^c = \text{softmax}(\boldsymbol{w}_c^\top \boldsymbol{H}_i^c), \quad i = 1, ..., L, \tag{2}$$

Where to look in an image

- obtain the object candidates.
- represent each candidate as a fix-length vector

$$\boldsymbol{o}^{spatial} = [\frac{x_{min}}{w_{img}}, \frac{y_{min}}{h_{img}}, \frac{x_{max}}{w_{img}}, \frac{y_{max}}{h_{img}}, \frac{x_{center}}{w_{img}}, \frac{y_{center}}{h_{img}}, \frac{w_{box}}{w_{img}}, \frac{h_{box}}{h_{img}}], \tag{3}$$

- take the output at the second fully-connected layer as the local information $\boldsymbol{o}^{local}$
- represent all the object candidates in an image by $\boldsymbol{O} = \{\boldsymbol{o}_1, \boldsymbol{o}_2, ..., \boldsymbol{o}_K\}$, where $\boldsymbol{o} = [\boldsymbol{o}^{local}, \boldsymbol{o}^{spatial}]$
- compute the attention on the i-th object.

$$\alpha_i^o = \text{softmax}(\boldsymbol{w}_o^\top \boldsymbol{H}_i^o), \quad i = 1, ..., K, \tag{4}$$

Accumulated Attention (A-ATT) model

Use the A-ATT model to combine multiple sources of information together, and generate a joint feature $\boldsymbol{H}$ to guide the assignment of attention?
For a feature sequence $\boldsymbol{X} = \{\boldsymbol{x}_1, ..., \boldsymbol{x}_n\}$,

$$\boldsymbol{H}_i = \tanh(\boldsymbol{W}_x \boldsymbol{x}_i + \boldsymbol{W}_{g_1} \boldsymbol{g}_1 + \boldsymbol{W}_{g_2} \boldsymbol{g}_2), \tag{5}$$

$$\tilde{\boldsymbol{x}} = \sum_{i=1}^{N} \alpha_i \boldsymbol{x}_i, \quad \alpha_i \in \{\alpha_i^q, \alpha_i^c, \alpha_i^o\} \tag{6}$$

Denote the process from Equation (5) to (6) as $\tilde{\boldsymbol{x}} = \text{ATT}(\boldsymbol{X}, \boldsymbol{g}_1, \boldsymbol{g}_2)$.

Accumulated Attention (A-ATT) model

For all three forms of information, their attended features can be represented as:

$$\text{A-ATT} = \begin{cases} \text{ATT}(\boldsymbol{S}, \tilde{\boldsymbol{o}}, \tilde{\boldsymbol{v}}) = \tilde{\boldsymbol{s}} \\ \text{ATT}(\boldsymbol{V}, \tilde{\boldsymbol{s}}, \tilde{\boldsymbol{o}}) = \tilde{\boldsymbol{v}} \\ \text{ATT}(\boldsymbol{O}, \tilde{\boldsymbol{v}}, \tilde{\boldsymbol{s}}) = \tilde{\boldsymbol{o}} \end{cases} \tag{7}$$

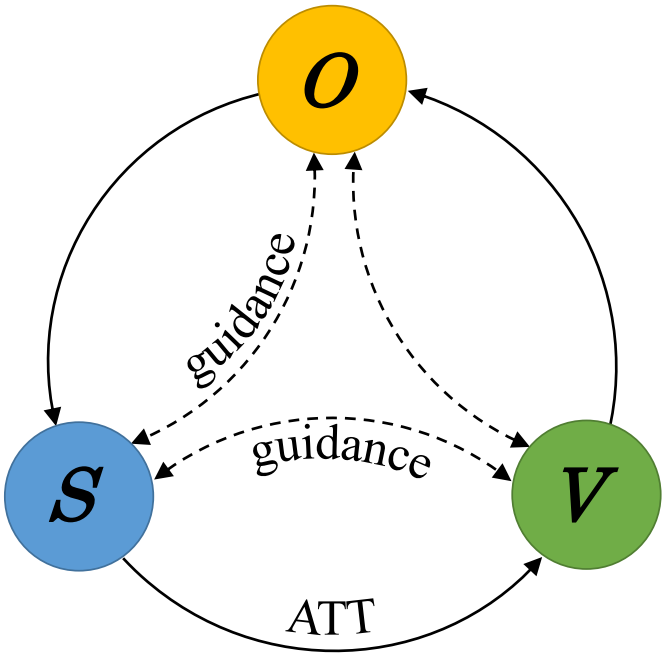
then we have

$$\text{A-ATT}_{i+1} = \mathcal{R}(\boldsymbol{S}, \boldsymbol{V}, \boldsymbol{O}, \text{A-ATT}_i),\ i \in \{1, 2, 3, \dots\}, \tag{8}$$

Accumulated Attention (A-ATT) model

Round	X	g_1	g_2	$\tilde{x}$
r1	S	$\mathbf{0}$	$\mathbf{0}$	$\tilde{s}_1$
	V	$\tilde{s}_1$	$\mathbf{0}$	$\tilde{v}_1$
	O	$\tilde{v}_1$	$\tilde{s}_1$	$\tilde{o}_1$
r2	S	$\tilde{o}_1$	$\tilde{v}_1$	$\tilde{s}_2$
	V	$\tilde{s}_2$	$\tilde{o}_1$	$\tilde{v}_2$
	O	$\tilde{v}_2$	$\tilde{s}_2$	$\tilde{o}_2$
...	...	...	...	...

Table: The process of A-ATT.



Grounding the query

- the probability for object $\boldsymbol{o}_i$ being the target object

$$\boldsymbol{F} = \tanh(\boldsymbol{W}_e[\tilde{\boldsymbol{s}}; \tilde{\boldsymbol{v}}]), \tag{9}$$

$$p_i = \text{softmax}(\boldsymbol{F} \odot \tanh(\alpha_i^o \boldsymbol{W} \boldsymbol{o}_i)), \tag{10}$$

- the final objective function

$$\begin{aligned} \mathcal{L} = &- \frac{1}{m} \sum_{j=1}^{m} \log p_j^* + \lambda_q \sum_{j}^{m} (1 - \boldsymbol{\alpha}^q)^2 \\ &+ \lambda_c \sum_{j}^{m} (1 - \boldsymbol{\alpha}^c)^2 + \lambda_o \sum_{j}^{m} (1 - \boldsymbol{\alpha}^o)^2, \end{aligned} \tag{11}$$

Experiment

Methods	ReferCOCO			ReferCOCO+			ReferCOCOg
	Val acc	TestA acc	TestB acc	Val acc	TestA acc	TestB acc	Val acc
Baseline	-	63.15	64.21	-	48.73	42.13	55.16
visdif	-	67.57	71.19	-	52.44	47.51	59.25
MMI	-	71.72	71.09	-	58.42	51.23	62.14
visdif+MMI	-	73.98	76.59	-	59.17	55.62	64.02
Luo *et al.*	-	74.14	71.46	-	59.87	54.35	63.39
Luo *et al.* (w2v)	-	74.04	73.43	-	60.26	55.03	65.36
Neg Bag	76.90	75.60	78.00	-	-	-	68.40
speaker+listener	77.84	77.50	79.31	60.97	62.85	58.58	72.75
speaker+listener+reinforcer	78.14	76.91	**80.10**	61.34	63.34	58.42	71.72
speaker+listener+MMI	78.42	78.45	79.94	61.48	62.14	58.91	72.13
speaker+listener+reinforcer+MMI	78.36	77.97	79.86	61.33	63.10	58.19	72.02
Ours-r2	80.68	81.37	79.79	65.35	68.36	60.19	72.67
Ours-r3	80.98	**81.67**	79.96	65.50	67.92	**60.69**	72.94
Ours-r4	**81.27**	81.17	80.01	**65.56**	**68.76**	60.63	**73.18**

Table: Comparisons on ReferCOCO, ReferCOCO+ and ReferCOCOg.

Experiments

Model	Train err	Val err	Test err
Human	9.0%	9.2%	9.2%
Random	82.9%	82.9%	82.9%
LSTM	27.9%	37.9%	38.7%
HRED	32.6%	38.2%	39.0%
LSTM+VGG	**26.1**%	38.5%	39.2%
HRED+VGG	27.4%	38.4%	39.6%
Ours-r2	29.3%	35.7%	36.5%
Ours-r3	30.5%	35.1%	35.8%
Ours-r4	29.8%	35.3%	36.3%
Ours-r3(w2v)	26.7%	**33.7**%	**34.2**%

Table: Comparisons on GuessWhat?!

Visualization

Figure: Attention on ReferCOCO

Visualization

Figure: Attention on ReferCOCO+

Visualization

Figure: Attention on ReferCOCOg

Visualization

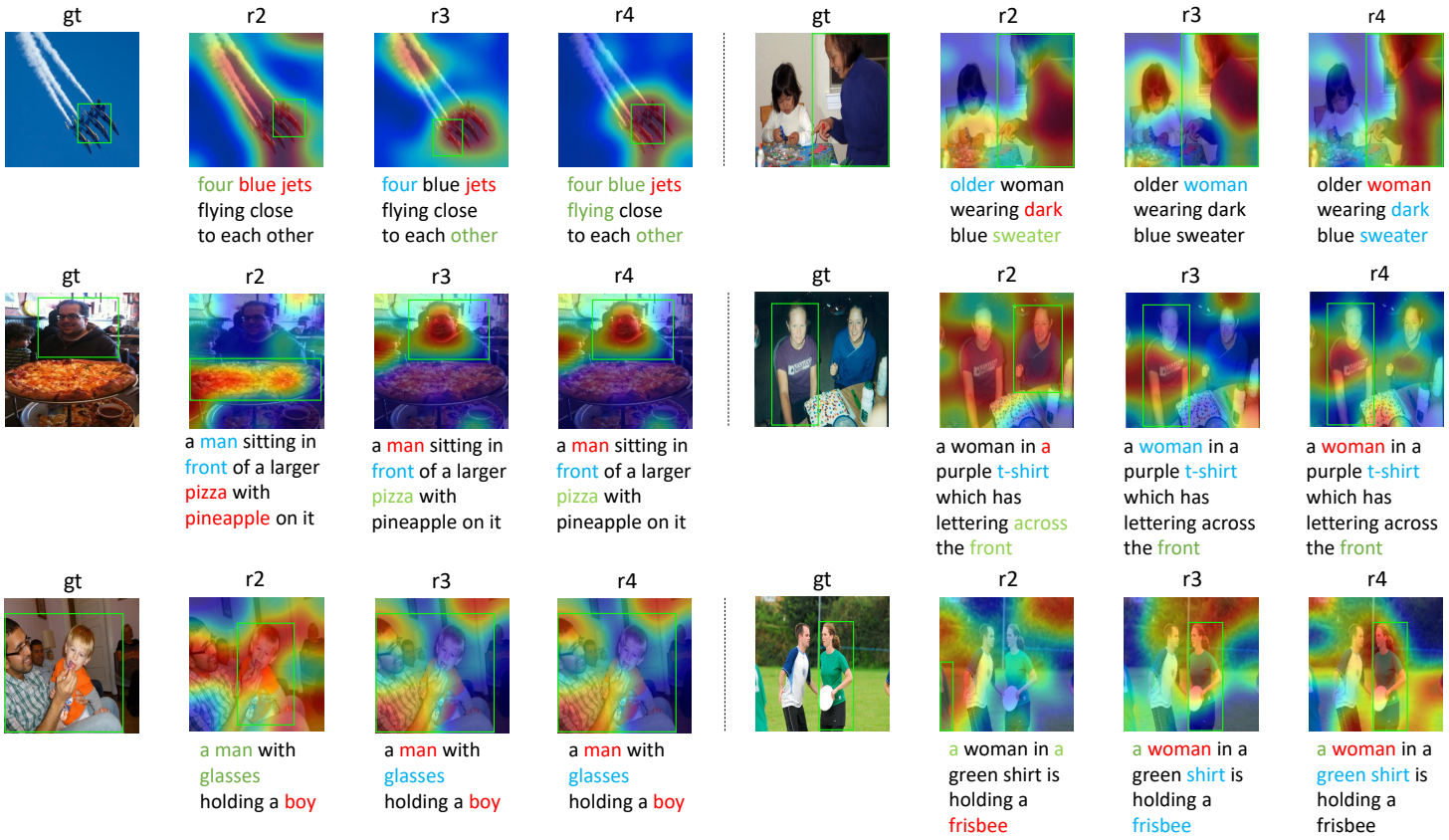


Figure: The process of attention accumulating.

THANK YOU